

Midterm Project 2:

INSTRUCTIONS:

Complete the following questions. Submit a word document with your name in the file name, and your RScript file, also with your name in the file name. This is an individual project. You can choose to use the variables I have suggested for the analyses, or you can choose to use different variables entirely. We have discussed multiple options in class.

DATASET INFORMATION:

This data approaches student achievement in secondary education of two Portuguese schools. The data attributes include student grades, demographic, social and school related features, and it was collected by using school reports and questionnaires. Two datasets are provided for two distinct subjects: Mathematics ([mat](#)

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) and Portuguese language ([por](#)

[Download por](#)

[Download por](#)[Download por](#)

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Code to merge the two datasets for students who appear in both as identified by searching for students with identical attributes in each is provided in the [student-merge.R](#)

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. In the merged dataset, columns with “.x” at the end of the column name come from the math dataset and columns with “.y” come from the Portuguese language dataset and is paired data for the students that appear there (each row is a unique student with their data from both the por and mat datasets. If the merge code is being problematic for you, the merge file is [here](#)

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A [text file](#)

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is provided that explains the meaning and values in each variable. Please note: the G1 and G2 values are the “quarter” grades that produce G3, the “semester” grade, and as such, G3 is highly correlated to both.

QUESTIONS: Create both a word document and an R script file for this project. Name the files with Project3_YourName.

For each of the following, state both research and null hypotheses, alpha, rejection criteria, whether the null was rejected and why, and state your conclusion as shown in class in words. Test all assumptions. Transform your data if necessary.

1. Chi-Squared Goodness of Fit Test: (40 pts) Gender and number of failures with expected frequencies being equal. (HINT: Sum the number of women who fail and the number of men who fail to form your contingency table). You may choose either dataset, or the merged dataset. If you choose the merged, only use one column (either the .y failures or the .x failures).
2. ANOVA: (40 pts) Choose 2 factors that you will test to see if they are predictive of G3 grades. You can use either data set or the merged dataset. (For example: weekday alcohol consumption and number of failures). Be sure to test for assumptions and test both the additive and interactive models. For any statistically significant results, perform a *post hoc* TukeyHSD and report your results as shown in class.
3. Choose an additional test to perform on this dataset. It could be a t-test (remember that the G1, G2 and G3 grades for mat or por are not independent, but G1 for por is independent of G1 for mat etc.), a correlation analysis or just another comparison using one of the tests above that you are curious about. (40 pts)